Parallel Seam Carving using OpenMP

Assignment 1 — High Performance Computing

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Exercise 1: We parallelize all algorithm steps with OpenMP, specifically using the basic parallel approach for cumulative energy and parallelizing only the copying phase of seam removal. All code is compiled with `-O3` optimizations.

Execution times are measured across five image resolutions (720×480 to 7680×4320) using 1, 2, 4, 8, 16, and 32 threads. Each test case requires at least 5 runs removing 128 seams to calculate average timings. Performance is evaluated via speed-up, defined as $S = \frac{t_s}{t_p}$, where t_s is the sequential runtime and t_p is the parallel runtime.

Exercise 2: We implement greedy seam removal to enable the parallel processing of multiple seams and we analyze the trade-offs between execution speed and final image quality. For each algorithm step and image resolution, we perform a scaling study to determine the optimal thread and core counts.

(a) We proceed now to implement triangle-based cumulative energy approach. We compared two tiling strategies where the image is divided into alternating upward and downward triangular blocks.

i. **Case 1 ($Y \pm 1$):** The tiles have a horizontal tip of two blocks. This approach present the pink dots not uniformly distributed.

ii. **Case 2 ($Y \pm 2$):** The tiles have a single-block tip. This approach present the pink dots uniformly distributed.

Computationally, both options present similar performance. However, in this study the second approach (single-block tip) seems more suitable because when the height of the triangle is $k = 1$, we fully return to the standard sequential approach, and, hence, we have a robust baseline to compare.

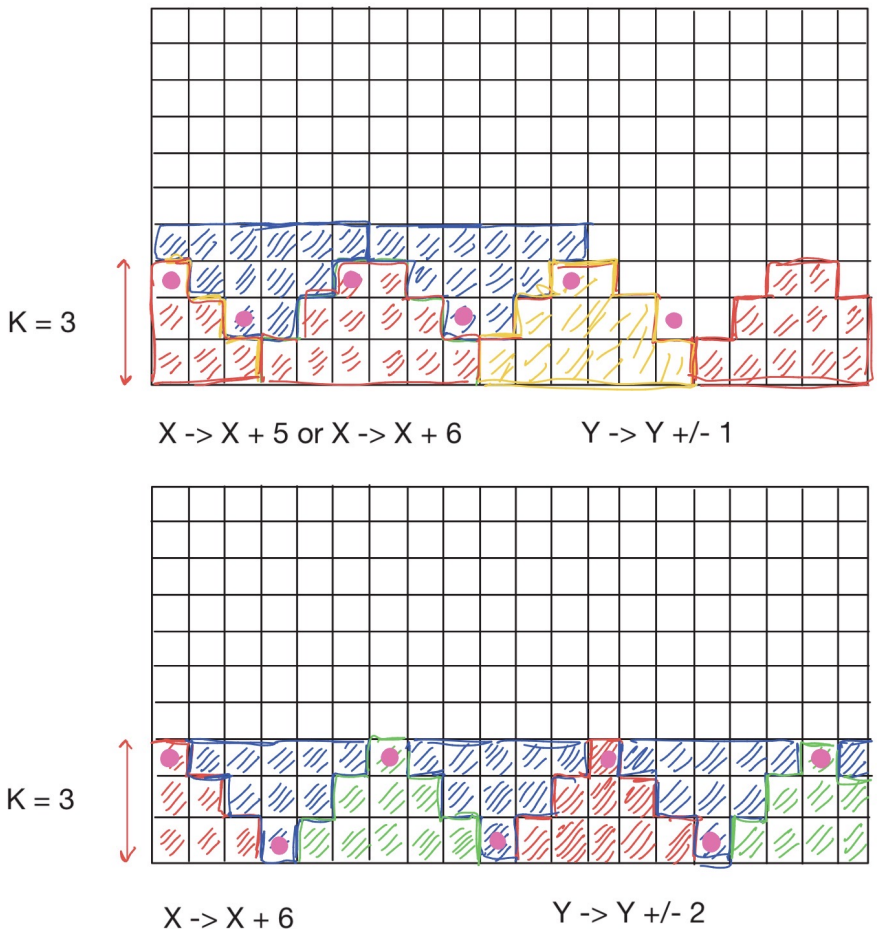

Figure 1: Comparison of wavefront tiling strategies for parallel seam carving. Case 1 (top) shows a non-uniform distribution of synchronization points (pink dots) due to a two-block tip. Case 2 (bottom) uses a single-block tip and uniform distribution of pink dots.

(b) To optimize the seam selection process, we modified the aforementioned traditional dynamic programming approach by implementing a bidirectional energy accumulation. The image is bisected at $\bar{y} = \lceil h/2 \rceil$, we execute the parallel triangular approach in parallel, on starting on the top and the other on the bottom as shown in image 2. The final seam is reconstructed by identifying the optimal midpoint $\hat{x}$ via the formula

$$\hat{x} = \underset{i}{\operatorname{argmin}} \left(E(x_i, \bar{y}) + \min \left(E(x_{i-1}, \bar{y}), E(x_i, \bar{y}), E(x_{i+1}, \bar{y}) \right) \right) \tag{1}$$

and performing two simultaneous tracebacks, one moving upward and one moving downward from the select $\hat{x}_i$.

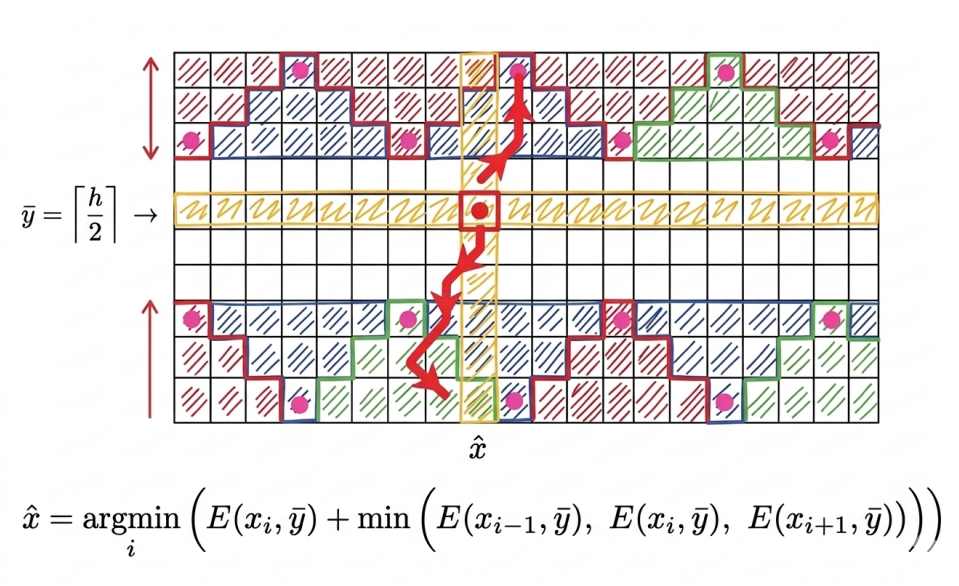

Figure 2: Bidirectional seam carving synchronization at the image midpoint $\bar{y}$. The upward and downward wavefronts (red arrows) converge at the central row, where the optimal transition point $\hat{x}$ is determined by minimizing the combined energy from both computed halves.